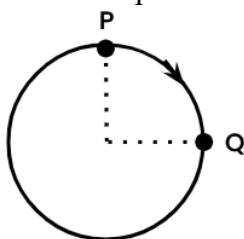


JEE-Main-04-04-2024 (Memory Based)
[EVENING SHIFT]

Physics

Question: A particle covers 90° along the circumference of a circle of radius 2 km as shown find its displacement.



Options:

- (a) 2 km
- (b) π km
- (c) $2\sqrt{2}$ km
- (d) 4 km

Answer: (c)

Question: 2 wires A and B are made of same material and have same mass. Radius of A is 2 mm and B is 4 mm. If resistance of B is 2Ω , resistance of A is

Options:

- (a) 4Ω
- (b) 8Ω
- (c) 16Ω
- (d) 32Ω

Answer: (c)

Question: In a system of 3 kg and 2 kg, 3 kg is moved by 2 units towards COM. How much should 2 kg move so that COM does not change position.

Options:

- (a) $\frac{4}{3}$ units
- (b) $\frac{5}{3}$ units
- (c) $\frac{7}{3}$ units
- (d) $\frac{7}{5}$ units

Answer: (a)

Question: A satellite revolves around the earth. The distance of the satellite at which it should be placed from the earth depends on time period of the earth, mass of the earth and universal gravitational constant

$$\{G = [M^{-1} L^3 T^{-2}]\}$$

Find the Relation between distance & Time Period

Options:

- (a) $R \propto T^3$
- (b) $R \propto T^{2/3}$
- (c) $R \propto T^{-3}$

(d) $R \propto T^{-2/3}$

Answer: (b)

Question: A gas of 1 mole at an initial temperature T expands adiabatically to double its volume. Find work done ($\gamma = 3/2$)

Options:

(a) $RT(2 - \sqrt{2})$

(b) $2RT$

(c) $RT/2$

(d) $3RT$

Answer: (a)

Question: A bar magnet has a magnetic moment of 0.5 Am^2 and it is placed in an external magnetic field of $8 \times 10^{-2} \text{ T}$. Find the work done when it is rotated from most stable position to unstable position

Options:

(a) MB

(b) $-MB$

(c) $2 MB$

(d) $-2 MB$

Answer: (d)

Question: In YDSE, Width of one slit is 4 times the other. Find ratio of maximum intensity to minimum intensity of the interference pattern on the screen

Options:

(a) $3/1$

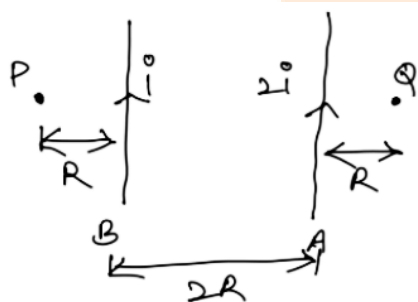
(b) $9/1$

(c) $4/1$

(d) $2/1$

Answer: (b)

Question: Two long wires A and B carry current i and $2i$ in the same direction as shown. Ratio of magnetic field at point P, at a distance of R on the right of B and at point Q, at a distance of R on the left of A, is $x/7$ then find x



Options:

Answer: (5)

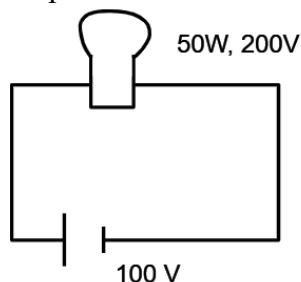
Question: A capacitor of capacitance 12.5 pF was charged by a voltage battery of 12 V . Then the battery is removed and a dielectric of dielectric constant 6 is introduced between the plates of the capacitors find the change in potential energy.

Options:

- (a) 775 pJ
- (b) -775 pJ
- (c) 900 pJ
- (d) -900 pJ

Answer: (b)

Question: A bulb rated 50W, 200V is connected to a battery of 100V. Find the power dissipated.



Options:

- (a) 50W
- (b) 25W
- (c) 12.5W
- (d) 6.25W

Answer: (c)

Question: Statement-1: Contact angle between liquid and solid depends on nature of the solid
Statement-2: Rise of a liquid in a capillary tube is independent of its inner radius

Options:

- (a) Only statement 1 is correct
- (b) Only statement 2 is correct
- (c) Both are correct
- (d) Both are incorrect

Answer: (a)

Question: A glass slab of refractive index $\sqrt{2}$ and thickness 3cm. Light is incident by a light at an angle equal to the critical angle of glass and air. Find lateral displacement of light ray as it comes out of the glass.

$$\sin 15^\circ = 0.25$$

Options:

- (a) $\sqrt{3}$ cm
- (b) $\sqrt{3}/2$ cm
- (c) $1/\sqrt{2}$ cm
- (d) 1 cm

Answer: (b)

Question: A charged particle of charge q is kept at the centre of the face of a cube. Find the flux linked with the cube

Options:

- (a) $\frac{q}{\epsilon_0}$
- (b) $\frac{q}{2\epsilon_0}$

- (c) $\frac{q}{6\epsilon_0}$
(d) $\frac{q}{24\epsilon_0}$

Answer: (b)

Question: Moment of momentum of an electron in the 4th orbit is

Options:

- (a) $\frac{h}{2\pi}$
(b) $\frac{h}{4\pi}$
(c) $\frac{2h}{\pi}$
(d) $\frac{h}{\pi}$

Answer: (c)



JEE-Main-04-04-2024 (Memory Based)
[EVENINGSHIFT]

Chemistry

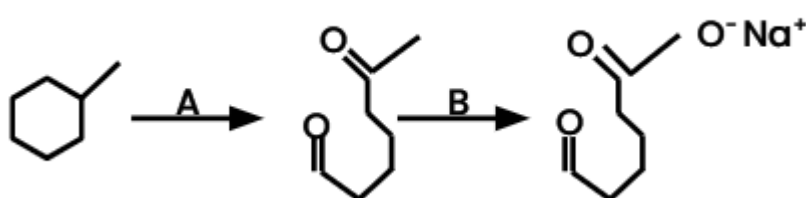
Question: Number of sigma and pi bonds in 2-oxo-hex-4-ynoic acid

Options:

- (a) 16
- (b) 18
- (c) 19
- (d) 17

Answer: (b)

Question: What is A & B ?



Options:

Answer: Solution : A- O_3 and Zn, CH_3COOH

B- NaOH, I_2

Question: Integer- 3 moles of gas is isothermal compressed from 60L to 40L through a constant pressure of 5 atm heat supplied is (-)_____

Options:

Answer: 100 L atm

Question: Vanillin from vanilla beans contains how many oxygen and pi bonds ?

Options:

- (a) 2-O and 2 pi bonds
- (b) 3-O and 4 pi bonds
- (c) 2-O and 3 pi bonds
- (d) 3-O and 2 pi bonds

Answer: (b)

Question: S - 1 : Correct order of ionization enthalpy is given as : $\text{F} > \text{Cl} > \text{Li} > \text{Na}$

S - 2 : Correct order of electron gain enthalpy is given as - $\Delta H_{\text{eq}} : \text{Cl} > \text{F} > \text{Li} > \text{Na}$

Options:

- (a) S1-Correct ; S2-Incorrect
- (b) S1-Incorrect ; S2-Correct
- (c) S1-Correct ; S2-Correct
- (d) S1-Incorrect ; S2-Incorrect

Answer: (c)

Question: Which of the following have a pyramidal shape ?

Options:

- (a) $\text{S}_2\text{O}_3^{2-}$
- (b) SO_4^{2-}
- (c) SO_3^{2-}
- (d) $\text{S}_2\text{O}_7^{2-}$

Answer: (c)

Question: IUPAC name of Catechol is :-

Options:

- (a) Benzene, 1, 2 - diol
- (b) Benzene - 1, 3 - diol
- (c) Benzene - 1, 4 - diol
- (d) 3 - Hydroxyphenol

Answer: (a)

Question: Consider the following statements :

Statement I : The number of emitted photoelectrons increases with increase in frequency of incident light.

Statement II : Kinetic energy of emitted photoelectrons increases with increase in frequency of incident light

Options:

- (a) Statement I is true but statement II is false
- (b) Statement I is false but statement II is true
- (c) Both Statement I and statement II are true
- (d) Both Statement I and statement II are false

Answer: (b)

Question: MnO^{2+} conc H_2SO_4 when reacted with which salt yield greenish yellow gas. Salt X contains :

Options:

- (a) F^-
- (b) Br^-
- (c) I^-
- (d) Cl^-

Answer: (b)

Question: Adsorbents used in adsorption chromatography :

Options:

- (a) Silica gel
- (b) Alumina
- (c) Both A and B
- (d) None of the above

Answer: (c)

Question: Find no. of compound with non zero dipole moment

BeCl_2 , BCl_3 , NF_3 , H_2O , H_2S , CCl_4 , XeF_4 , CO_2 , HBr

Options:

- (a) 4
- (b) 3
- (c) 5
- (d) 6

Answer: (a)

Question: Phthalimide $\xrightarrow[\text{(ii) Benzyl Chloride}]{\text{(i) KOH}}$
Find no. of π bonds in _____

Options:

Answer: ()

Question: The correct order of ionisation enthalpy for Li, Na, Cl, F is :

Options:

- (a) $\text{Na} < \text{Li} < \text{Cl} < \text{F}$
- (b) $\text{Li} < \text{Na} < \text{Cl} < \text{F}$
- (c) $\text{Na} < \text{Li} < \text{F} < \text{Cl}$
- (d) $\text{F} < \text{Cl} < \text{Li} < \text{Na}$

Answer: (a)

Question: Angular momentum for 4th orbit ?

Options:

- (a) h/π
- (b) $3h/2\pi$
- (c) $8h/\pi$
- (d) $2h/\pi$

Answer: (d)

Question: $\text{SO}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$; $K_c = 4.9 \times 10^{-2}$
 $2\text{SO}_3(\text{g}) \rightleftharpoons 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g})$ $K_c = ?$

Options:

- (a) 410
- (b) 420
- (c) 415
- (d) 425

Answer: (c)

Question: Which of the following statement is incorrect ?

Options:

- (a) Atoms are indivisible particles, which cannot be created or destroyed chemical reaction.
- (b) Atoms combine in any ratio to form compounds.
- (c) Atoms reorganize in a chemical reaction.
- (d) Atoms of different elements have different masses and chemical properties.

Answer: (b)

Question: Which of the following are the correct statements about fuel cell.

Options:

- (a) S - 1 : It is a galvanic cell
- (b) S - 2 Fuel cell have efficiency of 40%
- (c) S - 3 : It is eco friendly
- (d) S - 4 : Aluminium as a catalyst used in this cell

Answer: (d)

Question: Arrange the following in increasing order of their first ionisation enthalpy :

Al, Ga, In, Tl, B

Options:

- (a) $Tl < In < Ga < Al < B$
- (b) $In < Al < Ga < Tl < B$
- (c) $In < Ga < Al < B < Tl$
- (d) $B < Al < Ga < In < Tl$

Answer: (b)

Question: Correct order of stability

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (a)

Question: Find out the number of unpaired electrons in d-subshell for $[Co(H_2O)_6]^{3+}$

Options:

- (a) 3
- (b) 4
- (c) 0
- (d) 2

Answer: (c)

Question: Which of the following statement is incorrect ?

Options:

- (a) In homogeneous mixture composition is uniform
- (b) Compounds are formed when atoms of different element combine together in any ratio
- (c) Atoms of same element have identical atomic mass properties
- (d) In heterogeneous mixture composition is not uniform

Answer: (b)

Question: Match column I and column II

Column I	Column II
i) α - Glucose & α Galactose	a) Homologs
ii) α - Glucose & α - Fructose	b) Epimer
iii) α - Glucose & β - Glucose	c) Anomer
iv) α - Ribose & α - Glucose	d) Functional isomers

Options:

- (a) i - b, ii - d, iii - a, iv - c
- (b) i - b, ii - d, iii - c, iv - a
- (c) i - d, ii - b, iii - c, iv - a
- (d) i - a, ii - c, iii - d, iv - b

Answer: (b)

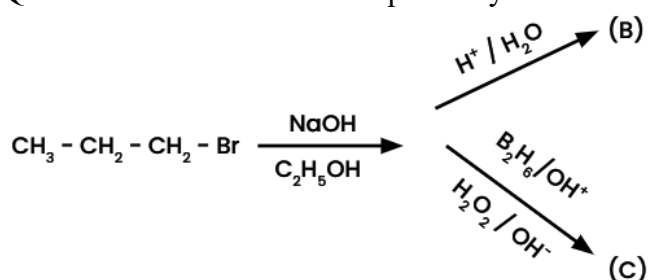
Question: Find the value of $x + y$ in given complex $[\text{Fe}(\text{NH}_3)_x(\text{CN})_y]^{-1}$

Options:

- (a) 6
- (b) 4
- (c) 5
- (d) 2

Answer: (a)

Question: What are B & C respectively ?



Options:

- (a) Propan - 1 - ol and propan - 2 - ol
- (b) Propan - 2 - ol and propan - 1 - ol
- (c) Both are propan - 1 - ol
- (d) Both are propan - 2 - ol

Answer: (b)

Question: Which of the following represents correct unit of slope of graph between molar conductivity (Δ_m) and $(\text{concn})^{1/2}$

Options:

- (a) $\text{S cm}^{1/2} \text{ mol}^{-1/2}$
- (b) $\text{S cm}^{3/2} \text{ mol}^{-2}$
- (c) $\text{S cm}^{7/2} \text{ mol}^{-3/2}$
- (d) $\text{S cm}^{5/2} \text{ mol}^{-3/2}$

Answer: (c)

Question: Calculate heat for isothermal process if expansion takes place from 20L to 60L against 5 atm pressure.

Options:

- (a) + 200
- (b) - 200
- (c) - 300
- (d) + 300

Answer: (a)

Question: S - 1 : Correct order of ionisation enthalpy is given as : $\text{F} > \text{Cl} > \text{Li} > \text{Na}$

S - 2 : Correct order of electron gain enthalpy is given as - $\Delta H_{\text{eq}} : \text{Cl} > \text{F} > \text{Li} > \text{Na}$

Options:

- (a) S1-Correct ; S2-Incorrect
- (b) S1-Incorrect ; S2-Correct
- (c) S1-Correct ; S2-Correct
- (d) S1-Incorrect ; S2-Incorrect

Answer: (c)

JEE-Main-04-04-2024 (Memory Based)
[EVENINGSHIFT]

Maths

Question: Team A has 4 men and 5 women. Team B has 4 women and 5 men. In how many ways can we pick 4 from each team such that there are 4 men and 4 women (integer type question)

Answer: 5626

Question: $\frac{1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + 100 \times 101^2}{1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \dots + 100^2 \times 101} =$

Options:

- (a) 305/302
- (b) 305/301
- (c) 301/305
- (d) 302/301

Answer: (b)

Question:

$$\int_{-1}^1 \frac{\cos ax}{(1+3^x)} dx = \frac{2}{\pi}$$

Find a

Options:

- (a) $\pi/4$
- (b) $\pi/3$
- (c) $\pi/2$
- (d) $\pi/5$

Answer: (c)

Question: Coeff of x^4, x^5, x^6 was in AP in expansion of $(1+x)^n$ Find maxm

Options:

- (a) 14
- (b) 7
- (c) 21
- (d) 0

Answer: (a)

Question: a, b, c in A.P., a, b, c AM was 8, find the cube of their GM such that a+1, b, c+3 were in GP

Options:

- (a) 120
- (b) 240
- (c) 360
- (d) 540

Answer: (a)

Question: If $\cos^{-1} x - \sin^{-1} y = \theta$ Find minimum of $x^2 + y^2 + 2xy \sin \theta$

Options:

- (a) 0
- (b) 1
- (c) 2
- (d) 3

Answer: (a)

Question: Let $y = y(x)$ be the solution of differential equation $(x^2 + 4)2dy + (2x^3y + 8xy - 2)dx = 0$ If $y(0) = 0$, then $y(2)$ is equal to.

Options:

- (a) $\pi/2$
- (b) $\pi/4$
- (c) $\pi/8$
- (d) $\pi/32$

Answer: (d)

Question: Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

and $B = I + \text{adj}(A) + \text{adj}(A)^2 + \text{adj}(A)^3 + \dots + \text{adj}(A)^{10}$. Then the sum

of all elements of the matrix B is -

Options:

- (a) 88
- (b) -88
- (c) 99
- (d) -99

Answer: (b)

Question: A parabola $y^2 = 12x$ has a chord PQ with PQ with mid-point (4, 1) then equation of PQ passes through:

Options:

- (a) $\left(\frac{1}{2}, -20\right)$
- (b) $\left(\frac{1}{2}, -10\right)$
- (c) $\left(-10, -\frac{1}{2}\right)$
- (d) $\left(-10, \frac{1}{2}\right)$

Answer: (a)

Question: The area (in sq. units) of the region described by $\{(x, y): y^2 \leq 2x, \text{ and } y \geq 4x - 1\}$ is

Options:

- (a) 9/31
- (b) 8/33
- (c) 9/32
- (d) 2/31

Answer: (c)

Question: A relation defined as $(x_1, y_1) R(x_2, y_2): x_1 \leq x_2 \text{ \& } y_1 \leq y_2$ and given that

- (a) R is reflexive but not symmetric
- (b) R is transitive . Then,

Options:

- (a) (a) is true and (b) is false
- (b) (a) is false and (b) is true
- (c) Both (a) and (b) are true
- (d) Both (a) and (b) are false

Answer: (c)

Question: Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and

$\vec{c} = x\hat{i} + 2\hat{j} + 3\hat{k}, x \in R$, If $\vec{d}$ is unit vector in the direction $\vec{b} + \vec{c}$ of such that $\vec{a} \cdot \vec{d} = 1$, then $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is equal to

Options:

- (a) 9
- (b) 10
- (c) 11
- (d) 12

Answer: (c)

Question: Let P be the POI of the lines, find the distance of P from the given plane

$$\frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1}, \frac{x-2}{2} = \frac{y-2}{3} = \frac{z-3}{2}$$

$$4x = 2y = z$$

Options:

- (a)
- (b)
- (c)
- (d)

Answer: (d)

Question: Let

$$f(x) = \int_0^x t + \sin(1 - e^t) dt, f(0) = 0, \text{ then } \lim_{x \rightarrow 0} \frac{f(x)}{x^3}$$

Options:

- (a) 1/6
- (b) -1/6
- (c) 1/2
- (d) -1/2

Answer: (b)

Question: If

$$f(x) = 3\sqrt{x-2} + \sqrt{4-x}$$

maximum value α and minimum value is β , then $\alpha^2 + \beta^2$

Options:

(a) 20

(b) 21

(c) 22

(d) 23

Answer: (c)

